

Integration Setup Guide

SharePoint & Database Connection — Step-by-Step

Before you start — what you need

Time required	15 min for SharePoint + 5 min for Database
Azure role	Application Administrator or Global Administrator
DB access	Database host, port, username, password, database name
SharePoint	Site URL: https://yourcompany.sharepoint.com/sites/yoursite

S P Part 1 — SharePoint Integration

NexusIQ uses Microsoft Graph API to search and retrieve documents from your SharePoint libraries. Follow these steps to register an Azure AD application and grant the required permissions.

1 Create an App Registration in Azure AD

Sign in to the Azure Portal at <https://portal.azure.com> using your Microsoft 365 admin account. Navigate to: **Azure Active Directory > App registrations > New registration**

Name : NexusIQ Supported account types: Accounts in this organizational directory only
Redirect URI : Single-page application (SPA) <http://localhost:3000>

Click Register. On the Overview page, copy the **Application (client) ID** and **Directory (tenant) ID**.

TIP These two IDs go into your frontend/.env file as VITE_AZURE_CLIENT_ID and VITE_AZURE_TENANT_ID, and into your backend .env as AZURE_CLIENT_ID and AZURE_TENANT_ID.

2

Add API Permissions

In your app registration go to: **API permissions > Add a permission > Microsoft Graph > Delegated permissions**

Search for and add each of these permissions:

User.Read Read user profile (sign-in) Sites.Read.All Read all SharePoint site collections Files.Read.All Read all files the user can access offline_access Refresh tokens without re-login

NOTE Sites.Read.All and Files.Read.All require admin consent. After adding all permissions, click "Grant admin consent for [Your Organisation]" and confirm.

TIP The status column shows a green checkmark once admin consent is granted successfully.

3

Create a Client Secret

Go to: **Certificates and secrets > Client secrets > New client secret**

Set a description (e.g. "NexusIQ") and expiry (24 months recommended). Click Add.

IMPORTANT Copy the secret Value immediately — it is only visible once. If you navigate away you must create a new secret.

Add to your backend .env file: `AZURE_CLIENT_SECRET=paste-your-secret-value-here`

4

Configure Redirect URI

Go to: **Authentication > Platform configurations**

Click **Add a platform** and select **Single-page application (SPA)**. Add these redirect URIs:

Development : `http://localhost:3000` Production : `https://yourdomain.com` (add when deploying)

Ensure **Implicit grant and hybrid flows** checkboxes are **UNCHECKED**. SPA mode uses PKCE — more secure than implicit flow.

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Find Your SharePoint Site ID

Method A — Graph Explorer (recommended):

```
1. Go to https://developer.microsoft.com/graph/graph-explorer 2. Sign in with your
Microsoft 365 account 3. Run: GET https://graph.microsoft.com/v1.0/sites?search=* 4.
Find your site and copy the full "id" field value
```

Method B — Direct API call:

```
GET https://graph.microsoft.com/v1.0/sites/yourcompany.sharepoint.com:/sites/yoursite
```

The Site ID has three parts separated by commas:

```
company.sharepoint.com,xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx,xxxxxxxx-xxxx-xxxx-xxxx-
xxxxxxxxxxxxxx
```

NOTE Copy the entire Site ID string including the commas. Add it to `SHAREPOINT_SITE_IDS` in your `.env` file.

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Update Your .env File

Open `~/Desktop/nexusiq/.env` and add or update these values:

```
AZURE_CLIENT_ID=your-application-client-id AZURE_TENANT_ID=your-directory-tenant-id
AZURE_CLIENT_SECRET=your-client-secret-value AZURE_REDIRECT_URI=http://localhost:3000
SHAREPOINT_SITE_IDS=company.sharepoint.com,siteId1,company.sharepoint.com,siteId2
```

Also update `frontend/.env`:

```
VITE_AZURE_CLIENT_ID=your-application-client-id
VITE_AZURE_TENANT_ID=your-directory-tenant-id
VITE_AZURE_REDIRECT_URI=http://localhost:3000
```

TIP Restart the SharePoint MCP server after updating `.env`: `python sharepoint_mcp.py`

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Connect SharePoint in NexusIQ

Open NexusIQ and go through the onboarding wizard or open **Settings (gear icon) > Connections > SharePoint tab**.

Paste your SharePoint site URL and click Connect with Microsoft. Document ingestion into ChromaDB starts automatically in the background.

TIP Monitor ingestion in the terminal: `python ingest.py`. A 339-page PDF typically takes about 30 seconds to ingest.

Part 2 — Database Integration

NexusIQ supports MySQL and Exasol databases, enabling natural language SQL queries against your data. You need database connection credentials from your administrator.

Database	Default Port	Driver	Best for
MySQL	3306	PyMySQL	General purpose OLTP workloads
Exasol	8563	pyodbc	Columnar analytics and data warehousing

1

Gather Your Database Credentials

You need the following from your database administrator or cloud console:

Host : Server hostname or IP (e.g. localhost, db.company.com, RDS endpoint) Port : 3306 for MySQL, 8563 for Exasol Username : A user with SELECT privileges on your database Password : The user password Database : Name of the database to connect to Schema : Schema name (same as database name for MySQL)

NOTE NexusIQ only requires SELECT (read-only) access. Avoid using a root or admin account in production.

2

Create a Read-Only User (Recommended)

For security, create a dedicated read-only database user for NexusIQ:

MySQL:

```
CREATE USER 'nexusiq'@'%' IDENTIFIED BY 'StrongPassword!'; GRANT SELECT ON your_database.* TO 'nexusiq'@'%'; FLUSH PRIVILEGES;
```

Exasol:

```
CREATE USER nexusiq IDENTIFIED BY 'StrongPassword!'; GRANT SELECT ON SCHEMA your_schema TO nexusiq;
```

TIP The NexusIQ SQL safety layer blocks all write operations (INSERT, UPDATE, DELETE, DROP etc.) regardless of user permissions.

3

Connect Your Database in NexusIQ

Open the onboarding wizard or go to **Settings > Database tab** in NexusIQ.

Select your database type, fill in the connection form, and click **Test Connection** first:

```
Database Type : MySQL or Exasol Host : your-db-host.com Port : 3306 (MySQL) or 8563
(Exasol) Username : nexusiQ Password : your-password Database Name : your_database
Schema : your_schema (optional)
```

A green "Connection successful" message confirms the credentials are correct. Click **Save and Connect**.

TIP The SQL MCP engine hot-swaps without requiring a server restart. The sidebar SQL MCP indicator turns green immediately.

4

Handle Special Characters in Passwords

The Settings UI handles URL encoding automatically. If editing .env manually, encode these special characters:

```
@ becomes %40 # becomes %23 % becomes %25 & becomes %26 + becomes %2B
```

Example .env DSN with encoded password (Pulsar@336 becomes Pulsar%40336):

```
DB_DSN=mysql+pymysql://nexusiQ:Pulsar%40336@localhost:3306/mydb
```

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Verify the Connection

After saving, the SQL MCP indicator in the NexusIQ sidebar should show a green "live" dot.

Test with natural language queries in the chat:

```
"What tables are in my database?" "How many customers do we have?" "Show me the top 5
products by revenue"
```

TIP NexusIQ automatically discovers your schema. Descriptive column names (e.g. customer_name vs col1) significantly improve query accuracy.

! Part 3 — Troubleshooting

Common issues and their fixes. For detailed errors check the FastAPI terminal log.

AADSTS70011 Invalid scope	Symptom: Microsoft sign-in fails with scope error	Fix: Add Sites.Read.All and Files.Read.All as Delegated permissions and grant admin consent in Azure AD.
Redirect URI mismatch	Symptom: Sign-in returns blank page or AADSTS50011 error	Fix: Verify the redirect URI in Azure AD Authentication matches exactly: <code>http://localhost:3000</code> — correct port, no trailing slash.
Admin consent required	Symptom: Permission denied accessing SharePoint documents	Fix: In Azure AD go to App registrations > API permissions and click "Grant admin consent". Requires Global Admin or Application Admin role.
SharePoint site not found	Symptom: Search returns no documents	Fix: Verify <code>SHAREPOINT_SITE_IDS</code> in <code>.env</code> . Use Graph Explorer to confirm the site ID. Format: <code>hostname,siteld,webId</code> (all three parts).
SQL connection refused	Symptom: Test connection fails or SQL MCP shows off	Fix: Check host, port, credentials. For MySQL ensure the user has privileges from your client IP: <code>GRANT SELECT ON db.* TO user@clientip</code> .
Special chars in DB password	Symptom: Connection fails with auth error despite correct password	Fix: URL-encode special characters in <code>.env</code> : <code>@</code> becomes <code>%40</code> , <code>#</code> becomes <code>%23</code> . The Settings UI does this automatically.
Documents not in RAG	Symptom: Questions about SharePoint docs return no results	Fix: Run: <code>python ingest.py --full</code> to force re-sync. Check that ChromaDB is running on port 8003. Supported types: <code>.pdf .docx .pptx</code>
Token expired	Symptom: SharePoint queries fail after initially working	Fix: MSAL auto-refreshes tokens. If issues persist, sign out and back in to NexusIQ to get a fresh Microsoft token.
MCP server shows off	Symptom: Sidebar shows SharePoint or SQL as off	Fix: Restart the relevant server: <code>python sharepoint_mcp.py</code> or <code>python sql_mcp.py</code> in a new terminal with the virtualenv activated.

Required Azure AD Permissions

Permission	Type	Required for	Admin consent required
User.Read	Delegated	Microsoft SSO sign-in	No
Sites.Read.All	Delegated	SharePoint document search	Yes
Files.Read.All	Delegated	Download file content	Yes
offline_access	Delegated	Refresh tokens automatically	No